

Companion LXIII: Persistent Homology of Weak-Lensing Fields in the Relaxation-Driven Cyclic Cosmology

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Abstract

Persistent homology provides a topological characterization of cosmological fields by tracking the birth and death of connected components, loops, and void-like structures across a continuous filtration. In weak-lensing analyses, persistence diagrams and barcodes quantify the stability of topological features in the convergence field. In the standard cosmological model, persistence reflects the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the stability, lifetime, and hierarchical merging of topological features in a characteristic way. This Companion develops a continuous description of persistent homology in RDCC, deriving how the relaxation scale influences persistence diagrams, barcode statistics, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive topological probe of the RDCC gravitational field.

1 Introduction

Weak-lensing convergence maps contain rich topological information. While Betti numbers count topological features at fixed thresholds, persistent homology tracks their evolution across a continuous filtration. This provides a powerful probe of non-Gaussianity, hierarchical structure, and the stability of topological features.

In the standard cosmological model, persistence diagrams reflect the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the birth, death, and lifetime of topological features in a scale-dependent manner, producing a distinctive signature in persistent-homology analyses.

2 Persistent Homology of the Convergence Field

For a 2D field such as the convergence $\kappa(\hat{\mathbf{n}})$, persistent homology tracks the evolution of topological features across a filtration parameter ν :

- β_0 features (connected components) are born when isolated high- κ regions appear.
- β_1 features (loops) are born when ring-like structures form.
- Features die when they merge into larger structures or collapse.

The persistence diagram consists of points $(\nu_{\text{birth}}, \nu_{\text{death}})$.

3 RDCC Modification of the Convergence Field

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (1)$$

The variance becomes

$$\sigma_{\kappa, \text{RDCC}}^2 = \sigma_{\kappa}^2 \left[1 - \chi_{\sigma} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

These modifications alter the birth and death thresholds of topological features.

4 Persistent Homology in RDCC

The RDCC persistence diagram becomes

$$D_{\text{RDCC}} = \left\{ (\nu_{\text{birth}}, \nu_{\text{death}}) \left[1 - \chi_D \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right] \right\}. \quad (3)$$

The lifetime of a feature is

$$L = \nu_{\text{death}} - \nu_{\text{birth}}, \quad (4)$$

and in RDCC,

$$L_{\text{RDCC}} = L \left[1 - \chi_L \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (5)$$

This modification reflects the relaxation-driven changes in topological stability.

5 Topological Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Persistent homology therefore exhibits:

- longer-lived large-scale connected components,
- suppressed short-lived small-scale features,
- fewer but more persistent loops,
- a shift in the persistence-lifetime distribution,
- a characteristic turnover near ℓ_{rel} ,
- modified hierarchical merging of topological structures.

These signatures provide a direct observational probe of the RDCC gravitational field in persistence space.

6 Conclusion

Persistent homology of weak-lensing fields in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the birth, death, and lifetime of topological features in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of topological cosmology, nonlinear structure, and the interplay between light and gravity in RDCC.

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